

Introduction to Social Data Analytics: Lecture 7 Part 1

Today's Lecture

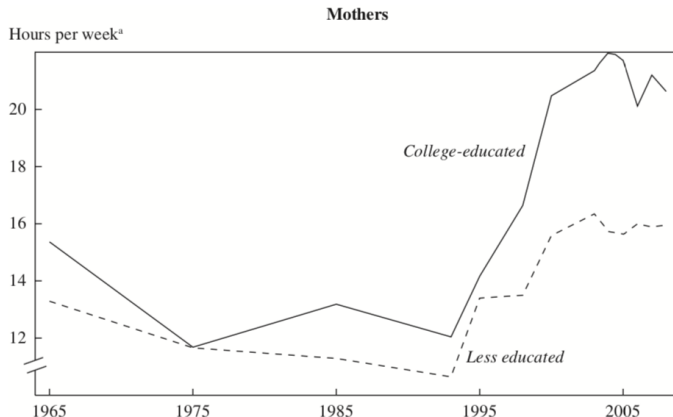
- ▶ Introduce the empirical application of the week: The Rug Rat Race by Ramey and Ramey (2010)
- ▶ Introduce loops
- ▶ Introduce the tidyverse and packages (more on this next time)

The Rug Rat Race (Ramey and Ramey 2010)

- ▶ Amount of time spent on childcare rose dramatically in 1990s
- ▶ Mothers in particular, college-educated parents in particular
- ▶ Why? Puzzling because these parents also had more to gain in the labor market

The Rug Rat Race (Ramey and Ramey 2010)

Figure 1. Time Spent on Childcare by Parents, by Educational Attainment, 1965–2008



- Before we discuss these results, we need to learn how to do some programming in R

for Loops in R

- ▶ The general form is:

```
for(some set of things){  
  do some stuff  
}
```

Writing each iteration out

```
#For i in 1:5, evaluate 2*i
```

```
i <- 1
```

```
2*i
```

```
## [1] 2
```

```
i <- 2
```

```
2*i
```

```
## [1] 4
```

```
i <- 3
```

```
2*i
```

```
## [1] 6
```

```
i <- 4
```

```
2*i
```

```
## [1] 8
```

Using a for loop is easier!

```
#One through Five  
for(i in 1:5){  
  print(2*i)  
}
```

```
## [1] 2  
## [1] 4  
## [1] 6  
## [1] 8  
## [1] 10
```

Using a for loop is easier!

#One through 100!

```
for(i in 1:100){  
  print(2*i)  
}
```

```
## [1] 2
```

```
## [1] 4
```

```
## [1] 6
```

```
## [1] 8
```

```
## [1] 10
```

```
## [1] 12
```

```
## [1] 14
```

```
## [1] 16
```

```
## [1] 18
```

```
## [1] 20
```

```
## [1] 22
```

```
## [1] 24
```

```
## [1] 26
```


You can iterate through any vector

```
for(i in c(3,10,99)){  
  print(2*i)  
}
```

```
## [1] 6
```

```
## [1] 20
```

```
## [1] 198
```

Back to the Rug Rat Race

- ▶ Next we are going to calculate mean time spent by mothers on childcare by year
- ▶ We will accomplish this using a forloop (there are easier ways to do this generally, which we will discuss soon)

Data

- ▶ The data comes from the American Time Use Survey which is a survey administered by the Bureau of Labor Statistics
- ▶ The data is based on “time diaries” which are detailed descriptions of the activities in a given day
- ▶ For example, if the interview was held on a Tuesday, the individual would report everything they did on Monday (from 4 AM Monday to 4 AM Tuesday)

Time Diaries

What did you do next? ← Question text

• **Read if necessary:** An activity is anything you did during the day. Activities include both active tasks like socializing, preparing food, or eating; and more quiet tasks like thinking and relaxing. Right now, you are talking to me on the telephone. Talking on the telephone is one type of activity. ← Interviewer Instructions

1. Sleeping

2. Grooming (self)

3. Watching TV

4. Working at main job

5. Working at other job

6. Preparing meals or snacks

7. Eating and drinking

8. Cleaning kitchen

9. Doing Laundry

10. Grocery shopping

11. Attending religious service

12. Paying household bills

30. Don't know/Can't remember

31. Refusal/ None of your business

← Pre-coded activities

	Start	ID	Activity	TIME	Hrs	Mins	Stop	Who	Who_2	Where	Where specify	← Variables
[1]	4:00AM		Sleeping	1	4		8:00AM					
[2]	8:00AM		Grooming	1		30	8:30AM					
[3]	8:30AM		Driving to work	1		30	9:00AM	0		12	Car, truck, or motor	
[4]	9:00AM		Working	1		30	9:30AM	0		2	Respondent's work	
[5]	9:30AM			1								
[6]												
[7]												
[8]												

Variables

- ▶ `dataset`
 - ▶ This variable indicates what year the data was collected
- ▶ `childdtot`
 - ▶ This variable reports the total number of hours spent on child care in a week

Our Steps

- ▶ Subset to just women with kids in household ages 25-34
- ▶ Create a list of years
- ▶ Iterate through the years
- ▶ Take the mean time spent on childcare for each year
- ▶ Print the output

Before getting to this, we are going to learn introduce the `tidyverse` package

Introducing the Tidyverse & Packages

- ▶ So far we have been using base R
 - ▶ But R is open source, meaning people can make packages in R
- ▶ This can make using R a lot easier
- ▶ Today we will download the tidyverse package, which is an incredibly useful package for data wrangling in R

Installing Package

- ▶ If you have never downloaded a package you must either navigate to the “Packages” tab and search for it
- ▶ You can also directly type `install.packages("packagename")` into the R console
- ▶ You only need to do this step once.

Loading a Package

- ▶ In order to use an installed package you need to load it into memory
- ▶ You do this by executing `library(packagename)`
 - ▶ You need to do this in every time you open R
 - ▶ For example, if you load “tidyverse” in homework 3 by executing `library(tidyverse)`, you will have to do this again for homework 4

Installing tidyverse

```
# If it not installed  
#install.packages("tidyverse")
```

```
# load into memory  
library(tidyverse)
```

```
## -- Attaching packages ----- tidyverse  
## v ggplot2 3.4.0      v purrr   0.3.5  
## v tibble  3.1.8      v dplyr  1.0.10  
## v tidyr   1.2.1      v stringr 1.4.1  
## v readr   2.1.3      v forcats 0.5.2  
## -- Conflicts ----- tidyverse_conflicts()  
## x dplyr::filter() masks stats::filter()  
## x dplyr::lag()    masks stats::lag()
```

Creating a tibble

```
rr <- read.csv("rugratrace.csv")
```

```
#Make a tibble
```

```
rr <- as_tibble(rr)
```

```
rr
```

```
## # A tibble: 106,020 x 54
```

```
##      state   age  sex ethnic under18 under5 ageyngst student act20 act21
##      <int> <int> <int>  <int>   <int>  <int>   <int>   <int>  <int>  <int>
```

```
## 1      23     26     2     -9         4        2      NA         0      20      40
```

```
## 2      23     53     2     -9         1        0      NA         0        0      10
```

```
## 3      23     21     2     -9         2        2      NA         0      20        0
```

```
## 4      23     24     2     -9         2        2      NA         0     335        0
```

```
## 5      23     25     2     -9         3        1      NA         0        5      40
```

```
## 6      23     33     2     -9         4        2      NA         0     173      60
```

```
## 7      23     26     2     -9         3        2      NA         0      24        2
```

```
## 8      23     61     1     -9         0        0      NA         0        0        0
```

Read in directly as a tibble

```
rr <- read_csv("rugratrace.csv")

## Rows: 106020 Columns: 54
## -- Column specification -----
## Delimiter: ","
## dbl (54): state, age, sex, ethnic, under18, under5, ageyngst, student, 
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this
```

The filter() functions

- ▶ The `filter()` functions selects rows of the data that meet a certain conditions
 - ▶ Ex: you want to restrict to individuals from a certain state, age or demographic
- ▶ The basic syntax is

```
filter(dataframe, some logical statement)
```

Where “some logical statement” is code that identifies which observations should be kept

Use filter to select rows

- ▶ Let's filter() to restrict to mothers aged 25-34 who have a child in the household

```
#Filter to mothers between 25-34, who have kids in the household  
filter(rr, age<=34 & age>=25 & mother2==1)
```

- ▶ This does not “save” the output. To save it you need to create an object

```
#You can create an object for this  
mothers2534 <- filter(rr, age<=34 & age>=25 & mother2==1)
```

Select in tidyverse

- ▶ `select()` chooses which variables you want to remain in your data
- ▶ For example, to select only the variables we require `dataset`, `mother2`, `age`, `childdtot` type:

```
#You can use "select" to select columns using variable names  
mothers2534 <- select(mothers2534, dataset, mother2, age, childdtot)
```

Get unique years

- ▶ Now let's loop through years and calculate the amount of time spent on childcare for each year
- ▶ Our first step is to create a vector of years using the `unique` function

```
years <- unique(mothers2534$dataset)
years
```

```
## [1] 1965 1975 1985 1993 1998 2003 2004 2005 2006 2007 2008
```


Let's try it first just setting i to 1

```
i <- 1  
sub <- subset(mothers2534, dataset==years[i])  
mean(sub$childtot)
```

```
## [1] 13.82249
```

Now let's include printing

```
i <- 1
#Subset mothers dataset just to year i
sub <- subset(mothers2534, dataset==years[i])
#Print out the year
cat("Year:", years[i], "\n")
```

```
## Year: 1965
```

```
#Calculate mean time spent on childcare for this group
cat("Average childcare per week:", mean(sub$childdtot), "\n")
```

```
## Average childcare per week: 13.82249
```

Let's put it together with the for loop

```
for(i in 1:length(years)){  
  #Subset mothers dataset just to year i  
  sub <- subset(mothers2534, dataset==years[i])  
  #Print out the year  
  cat("Year:", years[i], "\n")  
  #Calculate mean time spent on childcare for this group  
  cat("Average childcare per week:", mean(sub$childdtot), "\n")  
}
```

```
## Year: 1965  
## Average childcare per week: 13.82249  
## Year: 1975  
## Average childcare per week: 9.758393  
## Year: 1985  
## Average childcare per week: 12.86947  
## Year: 1993  
## Average childcare per week: 9.244037
```